+3.3V Boost Power Supply

FET RDS-ON
Find better parts

Battery reverse polarity protection

SW3 current rating would support ESP32 load. However, low cost PFET is more elegant design.

Q1 SI2301CDS-T1-GE3-ES

Q2 SI2301CDS-T1-GE3-ES

+BATT

J5 BATT_INPUT

2xAA Battery Input
3V Nominal
2V Low

BATT_EXT

BATT_PROTECTED

SW3

R4 10k

SHIELD

SS12F44G4

R22 10k

R23 49.9k

C28 1uF

BATT_VOLTAGE

+BATT is 3V Max
0.83x scale factor
to avoid A/D
non-linearity

L1 DFE322512F-4R7M=P2

D1 1N5819WS

U2 TPS61040DBV

VIN SW EN FB GND

1.233V

C4 4.7uF 16V

R3 169k

R2 100k

C7 68pF 50V

C5 4.7uF 16V

C6 4.7uF 16V

+3.3V

GND

LEDs

R6 470 — D2 XL-1608UBC-04 — Ice Blue

R21 470 — D3 XL-1608UYC-06FJ — Yellow

STATUS_LED

INT_LED_RED — R11 150 — 8mA

INT_LED_GREEN — R12 150 — 5mA

INT_LED_BLUE — R13 150 — 3mA

D4 NH-B2020RGBA-HF

2nd RGB
for rover battery?

Buttons

Flat

SW4 ZX-QC66-7.5TP — BTN0 — R5 10k — D5 3.3V — C9 0.1uF 50V

SW5 ZX-QC66-7.5TP — BTN1 — R7 10k — D6 3.3V — C10 0.1uF 50V

SW6 ZX-QC66-7.5TP — BTN2 — R8 10k — D7 3.3V — C11 0.1uF 50V

SW7 ZX-QC66-7.5TP — BTN3 — R9 10k — D8 3.3V — C12 0.1uF 50V

SW8 ZX-QC66-7.5TP — BTN4 — R10 10k — D9 3.3V — C13 0.1uF 50V

SW9 ZX-QC66-7.5TP — BTN5 — R14 10k — D10 3.3V — C14 0.1uF 50V

Joysticks

U3

LEFT_AXIS1 — R16 1k — C18 0.1uF 50V — D14 3.3V — R32 0

LEFT_AXIS2 — R15 10k — R33 0 — D13 3.3V — C17 0.1uF 50V

LEFT_BTN — R19 1k — D15 3.3V — C19 0.1uF 50V

U4

RIGHT_AXIS1 — R20 1k — C20 0.1uF 50V — D16 3.3V — R36 0

RIGHT_AXIS2 — R25 10k — R37 0 — D17 3.3V — C21 0.1uF 50V

RIGHT_BTN — R26 1k — D18 3.3V — C22 0.1uF 50V

Microcontroller

Identify board version by changing voltage divider. Increment in 0.1V steps.

V0.0 – DNU//10k, 0.000V
Future: 233k//10k, 0.097V

R34 330k — BOARD_VERSION — R35 10k

Pull-up needed to avoid getting stuck in code upload when powered from battery.

R24 10k — FLASH — SW2 FLASH — C8 0.1uF 50V

RESET — SW1 RESET — R1 10k — C2 1uF — C3 22uF 10V — C1 0.1uF 50V

U1 ESP32-S3-WROOM-1-N4

EN 3V3 GND PSRAM

IO0 IO1 IO2 IO3 IO4 IO5 IO6 IO7 IO8 IO9 IO10 IO11 IO12 IO13 IO14 IO15 IO16

TXD0 RXD0 IO17 IO18 USB_D- USB_D+ IO21 IO35 IO36 IO37 IO38 IO39 IO40 IO41 IO42 IO45 IO46 IO47 IO48

BATT_VOLTAGE
BOARD_VERSION
LEFT_AXIS1
LEFT_AXIS2
LEFT_BTN
RIGHT_AXIS1
RIGHT_AXIS2
RIGHT_BTN
BTN0 BTN1 BTN2 BTN3 BTN4 BTN5 BTN6 BTN7

UART_TXD0
UART_RXD0
PAIR_TXD1
PAIR_RXD1
USB_D-
USB_D+
STATUS_LED
INT_LED_RED
INT_LED_BLUE
INT_LED_GREEN

USB-C

J2 USB_C_Receptacle_USB2.0_16P

VBUS CC1 CC2 D- D- D+ D+ SBU1 SBU2 SHIELD GND

A4 A5 B5 A7 B7 A6 B6 A8 B8 S1 A1

LDO into +3.3V?

R28 0 — USB_D-

R29 0 — USB_D+

C23 4.7pF — C24 4.7pF — D21 3.3V — D22 3.3V

Find 5V TVS

R27 150 — R30 5.1k — R31 5.1k

PIN MUXING

Programming Connector

J1 PROG

1 UART_RXD0
2 UART_TXD0
3 FLASH
4 RESET
5 GND

Add Power

Pairing Connector

J3 PAIR

1 UART_RXD0
2 UART_TXD0
3 GND

D19 3.3V — D20 3.3V

IR Pairing?

Right Angle

SW10 TC-6615-8-260G — BTN6 — R17 10k — D11 3.3V — C15 0.1uF 50V

SW11 TC-6615-8-260G — BTN7 — R18 10k — D12 3.3V — C16 0.1uF 50V

ESD Diodes: SOT23-6 array packs?

T1 Tab T2 Tab T3 Tab T4 Tab T5 Tab

H1 MountingHole H2 MountingHole H3 MountingHole H4 MountingHole

Sheet: /
File: Rover_Remote.kicad_sch

Title: Rover Remote

Size: A3 | Date: 2025-12-09 | **Rev: 0**
KiCad E.D.A. 9.0.2 | Id: 1/1